

RaGeX Pricing Guardrail

Real-time guardrails and explanations for AI pricing decisions

A demo platform that detects risky AI price changes, estimates regret in ₹, and explains every decision using grounded policy evidence.

TEAM

Divija

Riya

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RaGeX CONSOLE

● API live

SKU-0001

HIGH RISK

₹313.40 → ₹563.58

Competitor ceiling breach · +79.8% move

EST. REGRET

₹1,562.21

EVIDENCE

4 snippets

ACTION

Hold

SKU-0117

Floor proximity

WARN

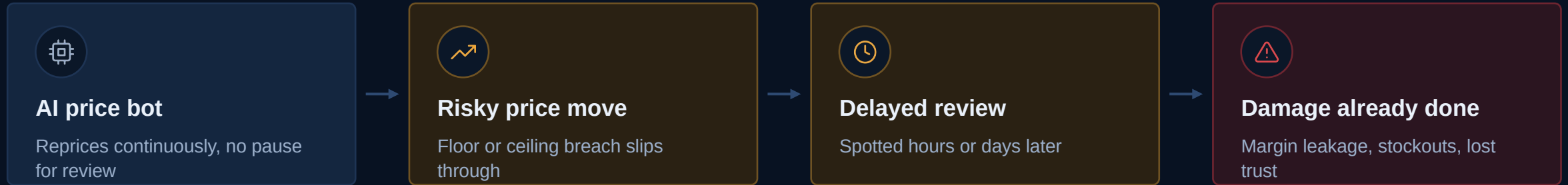
SKU-0042

Within policy band

PASS



AI pricing can move faster than human review



Who feels it first

- Pricing teams owning automated repricing
- Category managers defending margin targets
- Marketplace operators policing seller prices
- Retail and e-commerce automation teams

Where today's tooling stops

- Static rule checks with no context
- No counterfactual estimate of what the move costs
- Weak auditability once the price has shipped
- No policy-grounded explanation for the reviewer

Dashboards answer "what happened". They rarely explain why one decision is risky, or what it costs.

Real-time decision intelligence for pricing bots

- 
Capture
 Every pricing decision lands in one shared DecisionEnvelope JSON contract.
- 
Guard
 Live floor, ceiling, z-score spike and drift checks on every move.
- 
Score
 A scikit-learn reward model estimates counterfactual regret in ₹.
- 
Explain
 Answers grounded in the policy and incident corpus, with citations.
- 
Review
 One live dashboard for risk, regret, evidence and recommended action.

DECISION REVIEW

● live

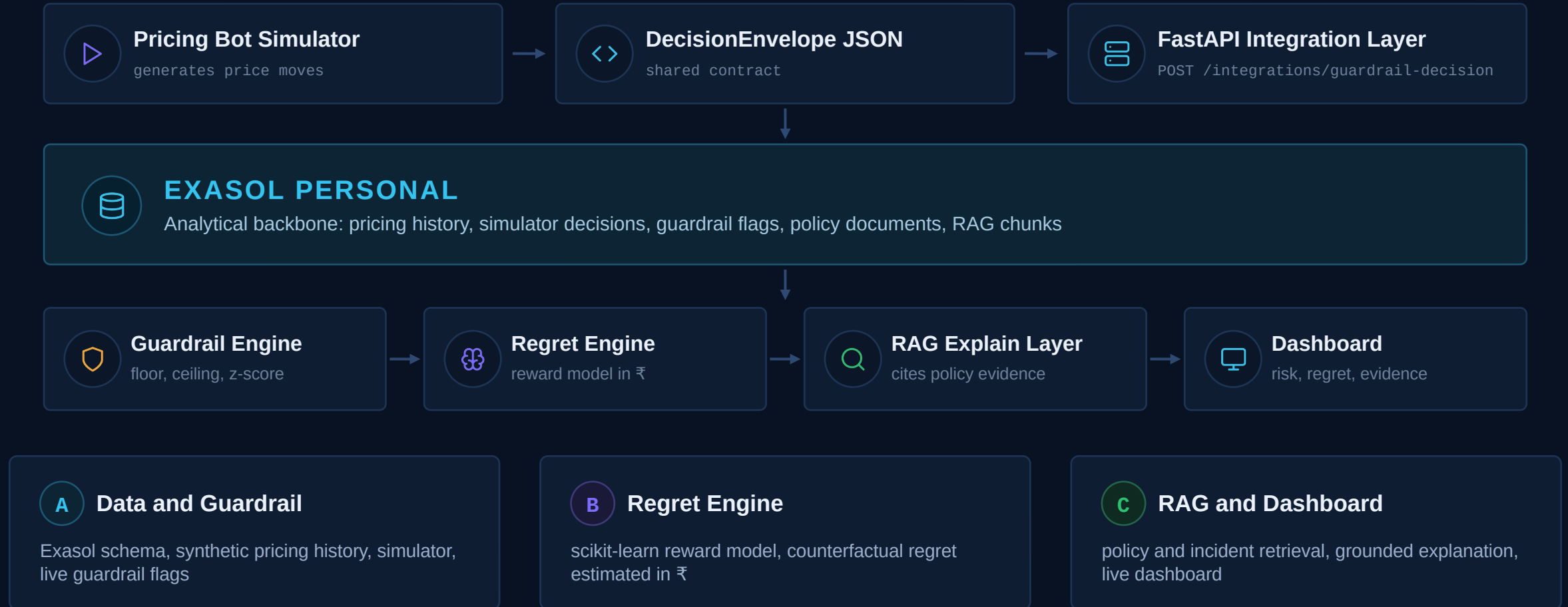
FLAGGED DECISIONS 12 of 40 scored	TOP REGRET ₹1,562 SKU-0001
EVIDENCE CITED 4 policy snippets	RECOMMENDATION Hold route to reviewer

GROUNDING EXPLANATION

Cites POL-03 §2.1 competitor ceiling rule and INC-2024-11 spike incident.

-  Rules, ML regret and RAG evidence in one path
-  Components joined only by contracts
-  Synthetic data, production-shaped path

Architecture



STACK FastAPI · Python · pandas · numpy · scikit-learn · Exasol Personal · PyExasol · RAG retrieval · Render-hosted demo

Why Exasol Personal matters



Role in the system

- Single analytical store for every pricing event and decision
- SQL inspection of anomaly patterns and decision history
- Realistic local analytical database, no enterprise infrastructure



Queries the demo runs

- Latest decisions by SKU
- Price movement patterns
- Full decision audit trail
- Flagged decision counts
- Historical z-score context
- RAG evidence retrieval

STORED IN EXASOL PERSONAL



EXASOL PERSONAL

HISTORICAL_PRICING_EVENTS

1,800 rows

MARKET_CONTEXT

1,800 rows

DECISIONS

simulator + scored

GUARDRAIL_FLAGS

live severity

KNOWLEDGE_CHUNKS

132 chunks

POLICY_DOCUMENTS

policies + incidents

1,800

seeded pricing decisions

1,800

market context rows

132

RAG knowledge chunks



integration endpoint tested



scored and explained end to end

Product Demo

RaGeX CONSOLE / DECISIONS

HIGH

WARN

PASS

● backend healthy

DECISION ID	SKU	PRICE MOVE	GUARDRAIL	REGRET
DEC-1042	SKU-0001	₹313.40 → ₹563.58	Competitor ceiling breach	HIGH ₹1,562.21
DEC-1039	SKU-0117	₹899.00 → ₹742.05	Floor proximity, 2.4σ	WARN ₹318.40
DEC-1036	SKU-0042	₹249.00 → ₹259.00	Within policy band	PASS ₹0.00

DECISION DETAIL · DEC-1042

₹313.40 → ₹563.58

SKU-0001 · flagged for competitor ceiling breach

ESTIMATED REGRET

₹1,562.21

GROUNDLED EXPLANATION

The new price sits above the competitor ceiling band defined in pricing policy, and matches a prior incident where a spike of this size triggered rollback.

EVIDENCE · 4 SNIPPETS

POL-03 \$2.1

POL-07 \$4.3

INC-2024-11

INC-2023-04

1 Flag detected

Guardrail marks the ceiling breach the moment the decision lands.

2 Regret estimated

Reward model converts the risky move into a ₹ figure.

3 Policy evidence cited

RAG layer returns snippets, and refuses unsupported questions.

4 Recommendation ready

Reviewer gets one action with the reasoning attached.

What we achieved

22

backend tests passing

1,800

synthetic history rows seeded

132

policy and incident chunks indexed



local Exasol Personal schema verified

● LIVE INTEGRATION ENDPOINT**POST /integrations/guardrail-decision**

accepts guardrail decisions from the simulator

**BEFORE**

Raw price change plus manual guesswork

- No severity signal on the change
- No estimate of what the move costs
- No evidence trail for the reviewer

**AFTER**

Flag, regret in ₹, evidence and recommendation

- Severity assigned the moment the decision lands
- Counterfactual regret quantified in ₹
- Policy snippets cited beside every verdict



Impact on the pricing review loop

- Cuts manual review time on flagged moves
- Catches risky price moves before damage compounds
- Gives reviewers evidence instead of black-box alerts
- Leaves an audit trail for pricing governance
- Answers one question: should this pricing decision be trusted?

Future Scope



Real pricing feeds

Connect to enterprise pricing and competitor data



Cloud Exasol backend

Move from Exasol Personal to a persistent SaaS deployment



Approval workflow

Reviewer accepts, overrides or rolls back a flagged price



Access and audit

Role-based access with immutable decision logs



Stronger reward model

Train on real outcomes with causal evaluation



Category risk view

Aggregate risk and regret above the SKU level



Severity alerting

Slack and email routing for high severity decisions



Policy simulation

A/B testing and what-if policy replay

RaGeX turns AI pricing from an opaque automation risk into a monitored, explainable and auditable decision system.

**"Every price decision should be explainable
before it costs money."**

Simulator



Exasol Personal



Guardrail



Regret Engine



RAG Explanation



Dashboard